

The Infinite-Bagged RSM Excess Risk R_k

Revised Exact Results and Rate Regimes

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Abstract

Under the common-loading forecast-error structure, the infinite-bagged RSM excess risk $R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}}$ is exactly the squared distance between the bagged weight vector and the oracle weight vector, measured in the error-covariance metric. This identity yields an exact projection/Jensen bracket for arbitrary positive precision profiles. A leading-order expansion of that exact bracket gives the open rate envelope $z_k^2 \lesssim R_k^{\text{bag},\infty} \lesssim z_k$, where $z_k = 1/k - 1/m$. After adding a finite-sample estimation penalty, this envelope implies that the optimal subset size lies between the cube-root and square-root regimes. The cube-root rule is obtained when precision heterogeneity is non-degenerate but bounded, so that no forecaster dominates. If one or a small group of forecasters carries a dominant share of total precision, the square-root boundary can reappear. Intermediate cases are best interpreted as a crossover between these two limits; a single exponent requires an additional local power-law or regular-variation approximation.

Reader's summary.

1. Under common loading, $R_k^{\text{bag},\infty} = \|\bar{v}_k - w^*\|_\tau^2$ exactly. The exact projection/Jensen bracket implies, to leading order, $z_k^2 \lesssim R_k^{\text{bag},\infty} \lesssim z_k$.
2. With the finite- T estimation penalty, the open envelope gives a tuning window between the cube-root and square-root regimes: roughly $T^{1/3} \lesssim k^* \lesssim T^{1/2}$.
3. If $0 < c \leq \tau_i/\bar{\tau} \leq C < \infty$ and $V_\delta \geq v > 0$, then $R_k^{\text{bag},\infty} \asymp z_k^2$, so $k^* \asymp T^{1/3}$.
4. If one or a few forecasters carry a non-vanishing share of total precision, bounded coherence fails; the z_k ceiling can become relevant and a square-root rule can reappear.
5. Intermediate cases are crossovers. A finite-range effective exponent $\alpha \in (1, 2)$ is useful, but a single scaling $k^* \asymp T^{1/(\alpha+1)}$ requires assuming that $R_k^{\text{bag},\infty}$ behaves locally like $C_R z_k^\alpha$.

1 Setup and notation

Assume the common-loading covariance of the forecast errors,

$$\Sigma_e = q\mathbf{u}\mathbf{u}' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2), \quad \tau_i = \sigma_i^{-2} > 0,$$

where $q \geq 0$ is the variance of the common unforecastable shock and τ_i is the idiosyncratic precision of forecaster i . Write

$$A_m = \sum_{i=1}^m \tau_i = m\bar{\tau}, \quad \ell_i = \frac{\tau_i}{A_m}, \quad A_S = \sum_{j \in S} \tau_j, \quad \langle a, b \rangle_\tau = \sum_i \frac{a_i b_i}{\tau_i}, \quad \|a\|_\tau^2 = \sum_i \frac{a_i^2}{\tau_i}.$$

RSM draws size- k subsets S uniformly without replacement, fits the constrained sum-to-one optimal weights within each subset, $v_{S,i} = \mathbf{1}\{i \in S\}\tau_i/A_S$, and bags. The infinite-bagging weight is $\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}]$, and the object of interest is $R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}} = \bar{v}_k' \Sigma_e \bar{v}_k - Q^{\text{opt}}$. We also use

$$V_\delta = \frac{1}{m} \sum_{i=1}^m \left(\frac{\tau_i}{\bar{\tau}} - 1 \right)^2, \quad H = \overline{\sigma^2} \bar{\tau} \geq 1, \quad z_k = \frac{1}{k} - \frac{1}{m}.$$

Here V_δ measures normalized precision dispersion, while H is the arithmetic-vs-harmonic gap; $H = 1$ iff all precisions are equal.

2 R_k as a squared weight deviation

Everything rests on one identity: $R_k^{\text{bag},\infty}$ is the squared distance of the infinite-bagged weight vector from the oracle weight vector.

Proposition 1 (Oracle weights). *The constrained-optimal full combination is $w_i^* = \ell_i = \tau_i/A_m$, with $Q^{\text{opt}} = q + 1/A_m$; for any subset S , $Q(S) = q + 1/A_S$.*

Proof. Since $\boldsymbol{\iota}' w = 1$, the common-loading term $w' q \boldsymbol{\iota}' w = q$ is constant. Thus minimizing $w' \Sigma_e w$ reduces to minimizing $\sum_i w_i^2 \sigma_i^2$ subject to $\sum_i w_i = 1$; the Lagrange solution gives $w_i \propto \tau_i$, and substituting yields $Q^{\text{opt}} = q + 1/A_m$. The subset result is the same argument restricted to S . $\square$

Proposition 2 (Exact quadratic-form identity). *Because w^* is the constrained minimizer and both $\bar{v}_k$ and w^* sum to one,*

$$R_k^{\text{bag},\infty} = (\bar{v}_k - w^*)' \Sigma_e (\bar{v}_k - w^*) = \|\bar{v}_k - w^*\|_\tau^2 = \sum_i \frac{(\bar{v}_{k,i} - \ell_i)^2}{\tau_i}.$$

Equivalently, with $g_i(k) = k \mathbb{E}_S[1/A_S \mid i \in S]$ and $\bar{v}_{k,i} = \tau_i g_i(k)/m$, $R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_i \tau_i (g_i(k) - 1/\bar{\tau})^2$.

Proof. Since $\Sigma_e w^* = (q + 1/A_m) \boldsymbol{\iota} = Q^{\text{opt}} \boldsymbol{\iota}$ and $\boldsymbol{\iota}' (\bar{v}_k - w^*) = 0$, the cross term vanishes and the common factor $q \boldsymbol{\iota}'$ drops. The final expression uses $\ell_i = \tau_i/A_m$ and the definition of $g_i(k)$. $\square$

Interpretation. At infinite bagging there is no Monte-Carlo selection variance; $R_k^{\text{bag},\infty}$ is the pure systematic distance between the bagged weights and the oracle weights. Increasing k moves the bagged weights toward the oracle, while finite- T estimation error pushes the other way by penalizing larger subsets.

Proposition 3 (Endpoints). *At $k = 1$, $\bar{v}_k = m^{-1} \boldsymbol{\iota}$, so R_1 is the excess risk of equal weights; at $k = m$, $\bar{v}_k = w^*$, so $R_m = 0$. Moreover $R_1 = Q^{\text{ave}} - Q^{\text{opt}} = \frac{H-1}{m\bar{\tau}}$, with $Q^{\text{ave}} = q + \frac{1}{m^2} \sum_i \sigma_i^2$.*

3 Exact bracket and leading-order envelope

Lemma 1 (Two exact moment identities). *For all k , $\sum_i \tau_i g_i(k) = m$ and $\sum_i g_i(k) = km \mathbb{E}_S[1/A_S]$.*

Proposition 4 (Exact projection/Jensen bracket). *For all $1 \leq k \leq m$,*

$$\frac{(k \mathbb{E}_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i \sigma_i^2 - m/\bar{\tau}} \leq R_k^{\text{bag},\infty} \leq \mathbb{E}_S[1/A_S] - \frac{1}{A_m}.$$

The denominator satisfies $\sum_i \sigma_i^2 - m/\bar{\tau} = m^2(Q^{\text{ave}} - Q^{\text{opt}}) \geq 0$, with equality only when all precisions are equal. The lower bound is a projection bound; the upper bound is the Jensen ceiling.

Proof. The upper bound is Jensen on $v \mapsto v' \Sigma_e v$: $Q_k^{\text{bag}, \infty} = (\mathbb{E}_S v_S)' \Sigma_e (\mathbb{E}_S v_S) \leq \mathbb{E}_S [v_S' \Sigma_e v_S] = Q_k^{\text{sub}}$; subtract Q^{opt} . For the floor, minimize $\sum_i \tau_i g_i^2$ over g satisfying the two identities of Lemma 1; the Gram/projection minimum, substituted into Proposition 2, gives the stated expression. $\square$

Equivalent projection form. Let $\delta_k = \bar{v}_k - w^*$ and $\eta = w^{\text{ave}} - w^*$. Then $R_k^{\text{bag}, \infty} = \|\delta_k\|_\tau^2$, $\|\eta\|_\tau^2 = Q^{\text{ave}} - Q^{\text{opt}}$, and the lower bound is the Cauchy-Schwarz projection $R_k^{\text{bag}, \infty} \geq \langle \delta_k, \eta \rangle_\tau^2 / \|\eta\|_\tau^2$. The exact floor uses only aggregate risks and the direction η ; the ceiling uses only the average-subset risk and is conservative because it averages risks rather than weights.

Corollary 1 (Leading-order open envelope). *Using $\mathbb{E}_S[1/A_S] = \frac{1}{k\bar{\tau}} + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + \dots$ with $\text{Var}_S(A_S) = k \frac{m-k}{m-1} \bar{\tau}^2 V_\delta$, the exact bracket implies*

$$\frac{V_\delta^2}{m(H-1)\bar{\tau}} z_k^2 \lesssim R_k^{\text{bag}, \infty} \lesssim \frac{1}{\bar{\tau}} z_k.$$

Remark 1 (What the envelope does and does not say). *The exact bracket is non-asymptotic under common loading; the z_k^2 - z_k expression is its leading-order rate implication. Since $0 < z_k < 1$, the envelope allows behavior between first and second order and does not by itself imply a single power law. It is safest to say the admissible effective exponent lies in $[1, 2]$; a literal $R_k^{\text{bag}, \infty} \asymp C_R z_k^\alpha$ requires an additional local power-law or regular-variation assumption.*

4 Finite- T loss and explicit boundary optimizers

Let $A = \sigma_\varepsilon^2 + Q^{\text{opt}}$ and use the idealized finite-sample loss $L_k = (A + R_k) \frac{T-1}{T-k}$. Here R_k becomes a tuning criterion: it decreases with k , while the inflation factor increases with k .

4.1 Upper-envelope boundary: square-root rule

Take the Jensen surrogate $R_k^U = C_U z_k$, $C_U = 1/\bar{\tau}$. Since $A = \sigma_\varepsilon^2 + q + C_U/m$, the loss becomes $L_U(k) = (A_0 + C_U/k) \frac{T-1}{T-k}$ with $A_0 = \sigma_\varepsilon^2 + q$. The continuous minimizer solves $A_0 k^2 + 2C_U k - C_U T = 0$, so $k_U^* = (-C_U + \sqrt{C_U^2 + A_0 C_U T})/A_0$ and, for large T , $k_U^* \sim \sqrt{C_U T/A_0}$.

4.2 Lower-envelope boundary: cube-root rule

Take the quadratic surrogate $R_k^L = C_L z_k^2$, $C_L = V_\delta^2/(m(H-1)\bar{\tau})$ (meaningful only when $H > 1$). The exact finite- m first-order condition is $-2C_L z_k \frac{T-k}{k^2} + A + C_L z_k^2 = 0$, equivalently k_L^* is the root in $(1, \min\{m, T\})$ of

$$(Am^2 + C_L)k^3 - 4C_L m k^2 + C_L(3m^2 + 2Tm)k - 2C_L T m^2 = 0.$$

For the interior case $k \ll m$ ($z_k \approx 1/k$), this reduces to $Ak^3 + 3C_L k - 2C_L T = 0$, with Cardano root

$$k_L^* = \sqrt[3]{\frac{C_L T}{A} + \sqrt{\left(\frac{C_L T}{A}\right)^2 + \left(\frac{C_L}{A}\right)^3}} + \sqrt[3]{\frac{C_L T}{A} - \sqrt{\left(\frac{C_L T}{A}\right)^2 + \left(\frac{C_L}{A}\right)^3}},$$

and, for large T , $k_L^* \sim (2C_L T/A)^{1/3}$.

4.3 General power surrogate

If $R_k \approx C_R z_k^\alpha$ with $\alpha \in [1, 2]$ and $k \ll m$, then $L_\alpha(k) \approx (A + C_R/k^\alpha)^{\frac{T-1}{T-k}}$, the first-order condition is $Ak^{\alpha+1} + (\alpha + 1)C_R k - \alpha C_R T = 0$, and $k_\alpha^* \sim (\alpha C_R T/A)^{1/(\alpha+1)}$. The envelope gives the window; it does not identify α .

4.4 Endogenous estimation and the square-root regime

The boundaries above describe the population object $R_k^{\text{bag}, \infty}$, in which the within-subset weights are the *known* optimal weights. In the endogenous case those weights are estimated by constrained OLS from T observations, which changes the tuning balance.

Two effects. (i) Fitting the $k - 1$ free weights of a size- k subset from T observations costs an estimation error of order $(k - 1)/T$, increasing in k — the classical Granger–Ramanathan / Elliott–Liao combination penalty, with k capping the number of estimated weights. (ii) The cube-root exponent $\alpha = 2$ relies on bagging the *exact* weights so that the first-order (Hájek) term cancels; estimation noise breaks that cancellation, so the systematic side reverts to the first-order average-subset suboptimality $Q_k^{\text{sub}} - Q^{\text{opt}} \asymp z_k \asymp 1/k$. The endogenous criterion is therefore, schematically,

$$L_k \approx A + \underbrace{\frac{c}{k}}_{\text{bias } \sim 1/k} + \underbrace{\frac{dk}{T}}_{\text{estimation}}, \quad -\frac{c}{k^2} + \frac{d}{T} = 0 \implies k^* \sim \sqrt{\frac{cT}{d}} \asymp \sqrt{T}.$$

The square root is recovered here not from leverage concentration (§5) but from estimation: the extra power of k in the cube-root bias is exactly the bagging cancellation that estimation destroys. The regime is governed by estimation severity m/T : for $m \ll T$ the cancellation largely survives and the cube root persists; for $m \gtrsim T$ (the empirically relevant case) the $(k - 1)/T$ penalty dominates and $k^* \asymp \sqrt{T}$. A Kan–Zhou-type correction $k_{\text{endo}}^* = k_{\text{exo}}^* \left(\frac{T-2}{T-m-2}\right)^{1/(\alpha+1)}$ interpolates and inflates k^* as $m \rightarrow T$.

A caveat on the covariance structure. The clean exogenous results — oracle weights $w_i^* = \tau_i/A_m$, the subset precision $A_S = \sum_{i \in S} \tau_i$, and the z_k closed forms — rely on common loading $\Sigma_e = \mathbf{q}\mathbf{u}' + D$ with D *diagonal*: a single common unforecastable shock plus idiosyncratic, mutually uncorrelated errors. For genuine external forecasters (e.g. SPF) this is defensible: the dominant cross-correlation is the common shock ε_t that every forecaster inherits, and the residual idiosyncratic errors are at least approximately uncorrelated. For endogenous, model-based forecasts the same structure is harder to justify: the competing models share data and overlapping predictors, so their errors around the conditional mean are themselves cross-correlated and D is far from diagonal. The realistic covariance is then a factor or banded (Toeplitz) structure rather than common loading — which is why the endogenous simulations use such covariances. Accordingly the endogenous $\sqrt{T}$ argument is best read as the *general* estimation-versus-suboptimality tradeoff — it needs only that fitting k weights costs $\sim k/T$ and that a k -subset combination is suboptimal at order $1/k$ — not as a consequence of the common-loading closed forms.

5 Two regimes and the crossover

The open envelope is two-sided but its exponents differ; which side is informative depends on the coherence of the precision profile.

Assumption 1 (Bounded coherence). *There exist $0 < c \leq C < \infty$, independent of m , with $c \leq \tau_i/\bar{\tau} \leq C$ for all i .*

The upper bound rules out a dominant forecaster; the lower bound rules out near-zero precisions and keeps the reciprocal moments of A_S controlled.

Assumption 2 (Non-degenerate heterogeneity). *There exists $v > 0$, independent of m , with $V_\delta \geq v$.*

This rules out the homogeneous case in which equal weighting is already near-optimal.

Theorem 1 (Cube-root regime). *Under Assumptions 1–2 there exist $0 < c_1 \leq c_2 < \infty$ depending only on c, C, v such that, for $2 \leq k < m$, $\frac{c_1}{m\bar{\tau}} z_k^2 \leq R_k^{\text{bag}, \infty} \leq \frac{c_2}{m\bar{\tau}} z_k^2$. Therefore $\alpha = 2$ and $k^* \asymp T^{1/3}$.*

Proof sketch. Bounded coherence controls the reciprocal expansion of $\mathbb{E}[1/A_S]$ and prevents a single coordinate from dominating the bagged-weight error; non-degenerate heterogeneity keeps the lower bound away from zero. The Jensen ceiling is thus pulled down to the same z_k^2 order as the projection floor, and $R_k \asymp C_R z_k^2$ in the finite- T loss gives the cube root. $\square$

Minimality. Each assumption protects one side:

Regime	Floor	Ceiling	Implication
bounded coherence + heterogeneity	z_k^2	z_k^2	$k^* \asymp T^{1/3}$
heterogeneity but no bounded coherence	z_k^2	z_k	k^* between $T^{1/3}$ and $T^{1/2}$
bounded coherence but no heterogeneity	0	z_k^2	averaging near-optimal
neither	0	z_k	no sharp rate

High coherence and dominant precision share. The square-root boundary is not caused by a merely negligible bad forecaster; it is caused by leverage concentration: one or a small group of forecasters carries a non-vanishing share of total precision, $\ell_{\max} = \max_i \tau_i / \sum_j \tau_j \not\rightarrow 0$, or a small group G has $\sum_{i \in G} \ell_i$ bounded away from zero while $|G|/m \rightarrow 0$. Then the oracle places substantial weight on a small set that random subsets include only intermittently, creating a slowly converging coordinate in $\bar{v}_k - w^*$, and the z_k ceiling can become the relevant boundary, so $k^* \asymp T^{1/2}$. The wording is “can reappear”, not “must reappear”, unless a precise leverage-concentration theorem is imposed.

Intermediate crossover. The intermediate case is a crossover, not a new exact power law. A useful phenomenological form is $R_k^{\text{bag}, \infty} \approx a z_k + b z_k^2$ ($a \geq 0, b > 0$), where $a z_k$ is coherence leakage and $b z_k^2$ the dispersion-driven bagging term. The local log-slope is

$$\alpha_{\text{eff}}(z_k) = \frac{d \log(a z_k + b z_k^2)}{d \log z_k} = \frac{a + 2b z_k}{a + b z_k} \in [1, 2].$$

A finite-range log–log fit can thus produce an effective exponent in $(1, 2)$; if the relevant range is well approximated by $R_k^{\text{bag}, \infty} \approx C_R z_k^\alpha$ then $k^* \asymp T^{1/(\alpha+1)}$, but that relies on local regular variation, not on the envelope alone.

6 Interpretation

One dial, two costs. The subset size k is a single bias–variance dial. At $k = 1$, RSM is equal weighting: no weight-estimation burden, but potentially large bias relative to the oracle if forecasters differ in precision. At $k = m$, RSM is full constrained OLS: no population weight bias, but maximal estimation burden. The optimum balances the declining systematic gap R_k against the increasing finite-sample penalty.

When equal weights are already good. If $V_\delta \approx 0$ and $H \approx 1$, then $R_1 = Q^{\text{ave}} - Q^{\text{opt}} \approx 0$: there is little to gain from estimating differentiated weights. Small k is appropriate not because estimation is easy, but because there is almost nothing useful to estimate.

When a few forecasts dominate. If a small set of forecasts carries most of the total precision, the oracle weights are far from equal, and the bagged weights need larger subsets to discover and up-weight those forecasts reliably. The convergence of the slowest important coordinate governs the rate, and the square-root boundary becomes relevant.

One good forecaster, or several? What matters is not the number of good forecasters but their share of total precision. If g forecasters carry essentially all precision, the coherence ratio behaves like $\kappa = \tau_{\max}/\bar{\tau} \approx m/g$. If $g \ll m$, coherence is high and the square-root boundary is plausible; if $g \propto m$, coherence remains bounded and the cube-root regime is the appropriate benchmark.

7 What to use in the paper

The safest paper-level statement is: under common loading, $R_k^{\text{bag},\infty}$ is exactly the squared covariance-metric distance between infinite-bagged RSM weights and oracle weights; the exact projection/Jensen bracket implies a leading-order envelope $z_k^2 \lesssim R_k^{\text{bag},\infty} \lesssim z_k$; therefore, after the finite-sample estimation penalty is added, the optimal subset size lies between the cube-root and square-root regimes. Under bounded coherence and non-degenerate heterogeneity the envelope closes at z_k^2 , giving $k^* \asymp T^{1/3}$. If a small set of forecasters carries a dominant share of total precision, the z_k ceiling can become relevant and the square-root boundary can reappear. In the endogenous case, estimating the within-subset weights reintroduces a first-order term and an $\mathcal{O}(k/T)$ penalty, so the square-root regime $k^* \asymp \sqrt{T}$ is the empirically relevant one when $m \gtrsim T$; this argument does not require the common-loading structure. Intermediate cases can be summarized by an effective exponent only as a local power-law approximation.